

CHRISTOPHER AL-RAHI

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EXPERIENCE

Curtiss-Wright Defense

Mechanical Engineering Intern

Ottawa, ON

September 2025 - December 2025

- Independently executed thermal redesign of liquid-cooled LFT modules using Simcenter FloTHERM, reworking internal coolant channel geometry to improve heat dissipation, and presented simulation results and design recommendations.
- Assembled and validated eight ruggedized electronic chassis for defense systems, ensuring compliance with military-grade reliability and quality standards.
- Performed environmental and thermal reliability testing on nine conduction-cooled products (4000+ insertions, 80+ thermal cycles) and reported product qualification and design verification.

CanmetENERGY - Renewable Heat & Power

Research Assistant

Ottawa, ON

May 2025 - August 2025

- Conducted root-cause analysis and restored the motion of a solar actuator system from $\pm 15^\circ$ to $\pm 40^\circ$, improving the tracking control and reliability of the system through mechanical redesign.
- Designed and implemented a height-adjustable shielded sensor system, increasing live temperature measurement accuracy by over 30%.
- Designed and built a compact, portable ventilation prototype for solar thermal testing, optimizing airflow and integration within tight spatial constraints.

CanmetENERGY - Alternative Energies Laboratory

Research Assistant

Ottawa, ON

January 2025 - April 2025

- Designed four ventilation and mechanical CAD layouts across three projects.
- Automated heat pump performance analysis with VBA, accelerating the process by 90%.
- Calculated heat and pressure losses to refine the mechanical design and improve system reliability.

PROJECTS

Inverted Pendulum SolidWorks, Arduino (C++), PID Control, 3D Printing

Designed every 3D printed part on a belt-driven cart that balances a rigid pendulum upright on a 0.5 m rail, using a rotary encoder to read pendulum angle with no cart position sensing. Sourced all the parts, optimized the electronics layout, and helped tune the PID controller that keeps the pendulum balanced and recovers it from disturbances.

Beer Goggles Arduino (C++), Load Cell + Ultrasonic Sensing, SolidWorks, 3D Printing

Led a team of five building a specific gravity and temperature monitoring system for Beyond the Pale Brewing that reads straight from the fermentation tank so the brewers do not have to sample by hand. Handled the mechanical integration, sensor calibration, and the Arduino code tying the system together. Read specific gravity within ± 0.1 SG across three rounds of testing.

Two-Stage Gear Reducer SolidWorks, Gear Design, Failure Analysis, Engineering Drawings

Designed a 9:1 two-stage gear reducer for an industrial conveyor drive that takes 6 hp at 400 rpm down to 45 rpm, built for a five-year service life. Sized the two-stage gear reduction, ran the bending and moment failure analysis, and designed every part in the assembly except the shaft.

EDUCATION

University of Ottawa

BASc. in Mechanical Engineering (Co-op)

Ottawa, ON

September 2022 - April 2027

AWARDS

Dean's Honour List | Recognition of Excellence

Awarded for maintaining a grade point average of 8.5 or better.

Faculty of Engineering

September 2022 and September 2024

SKILLS

Robotics & Control Systems:	PID Control, Motor Control, Sensor Fusion, Autonomous Systems
Embedded Systems & Hardware:	Arduino, Raspberry Pi, Sensors & Actuators, DAQ, IMU Integration
Mechanical Design & CAD:	SolidWorks, Creo, AutoCAD, GD&T, Engineering Drawings
Analysis & Prototyping:	FEA (ANSYS), Failure Analysis, Gear Design, 3D Printing
Programming & Automation:	Python, C++, VBA, Computer Vision (OpenCV)
Thermal & Testing:	FloTHERM, HVAC Design, Reliability & Environmental Testing